1. **You will find the syllabus of the course along with the course learning objectives on page 3.**
2. **MST : 25, EST: 45, Sessional: 10 (Project1) +10(Project2) + 5 (Certificate)+5 (Assignment)**
3. **MST topics are marked in yellow in the syllabus, a more detailed overview is provided here**

    a. Knowing your jetson kit architecture: You should be able to tell the types of jetson boards you are familiar with, difference between them, how do you choose, common components, how are they different from other embedded devices.
    b. NGC containers: How do you choose a particular container (version matching) for your task? Concept of Docker Image and Docker Container, adding, viewing, deleting them.
    c. Connecting your jetson, Running a docker container and the meaning of the arguments you provide while running
    d. Modifying and building your docker from scratch, what are the different ways in which you can add data/code/packages to your jetson
    e. Nvidia TAO toolkit: How to use it? What are the available containers.
    f. Inference: What are the different ways you can deploy your custom application (detection/classification/segmentation) ? How you can modify existing applications?
    g. Depth Estimation: You should be familiar with monocular camera calibration, stereo depth model, estimating depth using stereo correspondence.
    h. Optimization: You should be familiar with different model formats (tensorflow, pytorch, keras, onnx, torchscript etc) and their conversions along with config formats. You should be able to choose which format is most suitable in your case.

4. **Sample questions:-**

    a. You will be given any sample application and you need to write down how you will proceed with the deployment. Your answer should include whether you will use a standard docker or build one from scratch, in both cases you need to say how you will proceed i.e. steps to build or run existing. In all cases your answer should clearly explain your choice, all parameters you are using. Further you will start with the particular detection/classification/segmentation models you will use. You should be able to provide the names and the way in which you will run them. While model architecture details are not required, you need to write code snippets where you will mention the custom changes you are making to the existing models.
    b. If your application requires depth estimation along with detection/segmentation, you will need to explain calibration steps and depth estimation (code snippets)
    c. You need to be familiar with the tone of the question. For example, if the application required depth information, your answer should focus on estimation using code. However, if the question is more towards designing, you may need to give the mathematical model details.
    d. If you are told you have a trained model but you need to optimize it, you should be ready with conversion modules between torch and onnx, torch and torch script etc. and explain your choice.
    e. Trouble shooting: You may be given a scenario and you may need to analyse the problem. What possibly did you do wrong?

UCS671- Edge AI and Robotics: Embedded Vision

Course Objective: This course will provide students with advanced conceptual knowledge and practicals on various computer vision and deep learning applications and provide the overall environment for end-to-end pipeline development from data collection to deployment.

Introduction: Utilizing Jetpack SDK and other NVIDIA Toolkits to deploy CNN models on Jetson, Creating Jetbot kits and deploying various applications, Working with NVIDIA Robotics toolkit: Isaac SIM SDK and Gazebo for collision avoidance, path following.

Introduction to Edge AI: AI at the Edge & IoT, Jetson Architecture, Getting Started with Jetpack, NGC Containers in Jetson, Getting started with NGC & Containers on Jetson.

Introduction to NVIDIA Toolkits and SDKs: Transfer Learning Toolkit, TensorRT Optimization, Triton Inference Server for Inferencing and Deployments, Various Visualization Tools, Kubernetes Deployment, Deepstream SDK, Deploying Classification, Detection and Segmentation CNN Models on Jetson Devices.

Perception & Autonomous Navigation: Building JetBot Kits, Introduction to basic motion on JetBot, Collision Avoidance: Stop/Go classifier (JetBot), freespace detection, Path Following: Recording user input/video + DNN training (DriveNet), Simulation: Gazebo & Isaac SIM.

Advanced Vision & SLAM: Pose Recognition (Deploying Human pose model), Depth Estimation: Mono/Stereo depth and point extraction, Visual Odometry: Camera pose estimation from DNNs, SLAM on JetBot.

Practical Labs:

- Setting up the Jetson Project kit.
  {DLI Online Course: Getting Started with AI on Jetson Nano.}
- Deployment of Various Classification, Object Detection and Segmentation models using TensorRT, and Triton Inference Server in Jetson Nano.
- Getting started building various Jetbot Kits.
- Basic Motion with Jetbot
- Collision Avoidance with Jetbot kit
- Object following and Road following (DriveNet) with Jetbot.
- Teleoperation with Jetbot.
- Human Pose Estimation in Jetson Nano/JetBot.
- Implementing SLAM on Jetbot.

Course Learning Outcomes (CLO) / Course Objective (CO):

After the completion of this course, the students will be able to:

1. Introducing concepts of Edge computing and Jetson kits.
2. Provide understanding of approaches, concepts and algorithms used in computer vision deployments.
1. Practice implementation of various concepts and tools covered in the course.
2. Utilize programming and scientific tools for relevant software implementation.
3. Understand GPU computing for building advanced computer vision pipelines on Jetson devices.

Text Books:

1. Computer Vision: Algorithms and Applications, R. Szeliski, Springer, 2011.
2. Computer Vision: A Modern Approach, D. Forsyth and J. Ponce, Prentice Hall, 2nd ed., 2011.
3. Richard Szeliski, Computer Vision: Algorithms and Applications, Springer-Verlag London Limited 2011.

Reference Books:

1. Richard Hartley and Andrew Zisserman, Multiple View Geometry in Computer Vision, Second Edition, Cambridge University Press, March 2004.
2. K. Fukunaga; Introduction to Statistical Pattern Recognition, Second Edition, Academic Press, Morgan Kaufmann, 1990.
3. R.C. Gonzalez and R.E. Woods, Digital Image Processing, Addison- Wesley, 1992.
4. Christopher M. Bishop; Pattern Recognition and Machine Learning, Springer, 2006.